Lab 4

Task 2.2 $\int_0^\infty h(t) e^{-j\omega t} dt$

$$h(t) = \frac{G}{\tau_1 - \tau_2}\left(e^{-t/\tau_1} - e^{-t/\tau_2}\right) u(t)$$

$$H(j\omega) = \frac{G}{\tau_1 - \tau_2}\left[\int_0^\infty e^{-t/\tau_1} e^{-j\omega t} dt - \int_0^\infty e^{-t/\tau_2} e^{-j\omega t} dt\right]$$

$$H(j\omega) = \frac{G}{\tau_1 - \tau_2}\int_0^\infty e^{-(1/\tau_1 + j\omega)t} dt - \int_0^\infty e^{-(1/\tau_2 + j\omega)t} dt$$

$$I_1 = \frac{-e^{-(1/\tau_1 + j\omega)t}}{1/\tau_1 + j\omega}\Bigg|_0^\infty = 0 - \frac{(-1)}{1/\tau_1 + j\omega} = \frac{\tau_1}{1 + j\omega\tau_1}$$

So I_2 must be $\frac{\tau_2}{1 + j\omega\tau_2}$

$$H(j\omega) = \frac{G}{\tau_1 - \tau_2}\left[\frac{\tau_1}{1 + j\omega\tau_1} - \frac{\tau_2}{1 + j\omega\tau_2}\right] \rightarrow \text{cross multiply}$$

$$H(j\omega) = \frac{G}{\tau_1 - \tau_2}\,\frac{\tau_1(1 + j\omega\tau_2) - \tau_2(1 + j\omega\tau_1)}{(1 + j\omega\tau_2)(1 + j\omega\tau_2)}$$

Numerator $\tau_1 + j\omega\tau_2\tau_1 - \tau_2 - j\omega\tau_1\tau_2 = \tau_1 - \tau_2$

$$H(j\omega) = \frac{G}{\tau_1 - \tau_2} \cdot \frac{\tau_1 - \tau_2}{(1 + j\omega\tau_1)(1 + j\omega\tau_2)} = \frac{G}{(1 + j\omega\tau_1)(1 + j\omega\tau_2)}$$

$$|H(j\omega)| = \frac{G}{\sqrt{1 + (\omega\tau_1)^2}\sqrt{1 + (\omega\tau_2)^2}}$$

$$\angle H(j\omega) = -\tan(\omega\tau_1) - \tan(\omega\tau_2)$$